Radix Sparse Merkle Trees

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1 Definitions

Definition 1 (Tree) *A tree T is either an empty set $\emptyset$ or a triple $(T_0, T_1; c)$ where T_0, T_1 are trees (called the* left *and the* right subtree *of T respectively) and c is data of any type (called the* root data*). A tree of the form $(\emptyset, \emptyset; c)$ is called a* leaf*.*

Definition 2 (Left and right subsets) *For any non-empty $K \subseteq \{0,1\}^\kappa$ with $\mathsf{gcp}(K) = p$, the sets $K_0 = \{k \in K\colon p\|0 \preceq k\}$ and $K_1 = \{k \in K\colon p\|1 \preceq k\}$ are called the left and the right subset, respectively. Note that $K_0 \cap K_1 = \emptyset$, $|K_0| < |K|$, and $|K_1| < |K|$.*

Definition 3 (Left and right functions) *For any non-empty $K \subseteq \{0,1\}^\kappa$ with $\mathsf{gcp}(K) = p$, and a function $R\colon K \to V$ the restrictions $R_0 = R|_{K_0}$ and $R_1 = R|_{K_1}$ are called the left and the right function, respectively.*

Definition 4 (Radix tree) *The* radix tree *$T = \mathsf{RT}(K, R)$ for a set $K \subseteq \{0,1\}^\kappa$ and a function $R\colon K \to V$ is defined as follows:*

1. *If $K = \emptyset$ then $\mathsf{RT}(K, R) = \emptyset$*
2. *If $K \neq \emptyset$ then $\mathsf{RT}(K, R) = (\mathsf{RT}(K_0, R_0), \mathsf{RT}(K_1, R_1); p, w)$, where $p = \mathsf{gcp}(K)$ (also denoted by $\mathsf{gcp}(T)$) and $w = R(p)$ if $p \in \{0,1\}^\kappa$ and $w = \perp$ otherwise. Note that $p \in \{0,1\}^\kappa$ if and only if $K = \{p\}$ and $K_0 = K_1 = \emptyset$.*

Definition 5 (Adding an element) *Adding a key-value pair $(k, v) \in \{0,1\}^\kappa \times V$ to a radix tree is defined as follows: (1) $\mathsf{Add}(\emptyset; k, v) = (\emptyset, \emptyset; k, v)$, (2) $\mathsf{Add}((\emptyset, \emptyset; k, v'); k, v) = (\emptyset, \emptyset; k, v')$, and (3) if $T = (T_0, T_1; p, w)$:*

1. *$\mathsf{Add}(T; k, v) = ((\emptyset, \emptyset; k, v), T; \mathsf{gcp}(p, k), \perp)$ if $p \not\preceq k$ and $\mathsf{gcp}(p, k)\|0 \preceq k$*
2. *$\mathsf{Add}(T; k, v) = (T, (\emptyset, \emptyset; k, v); \mathsf{gcp}(p, k), \perp)$ if $p \not\preceq k$, and $\mathsf{gcp}(p, k)\|1 \preceq k$*
3. *$\mathsf{Add}(T; k, v) = (\mathsf{Add}(T_0; k, v), T_1; p, w)$ if $p\|0 \preceq k$*
4. *$\mathsf{Add}(T; k, v) = (T_0, \mathsf{Add}(T_1; k, v); p, w)$ if $p\|1 \preceq k$*

2 Canonicity of Addition

Lemma 1 $\mathsf{Add}(\mathsf{RT}(K,R);k,v) = \mathsf{RT}(K\cup\{k\}, R\cup\{(k,v)\})$ *for every* $K \subseteq \{0,1\}^\kappa$, $R\colon K \to V$, $k \in \{0,1\}^\kappa\backslash K$, *and* $v \in V$.

Proof: Note that $\mathsf{Add}(\mathsf{RT}(\emptyset,R);k,v) = \mathsf{Add}(\emptyset;k,v) = (\emptyset,\emptyset;k,v) = \mathsf{RT}(\{k\},\{(k,v)\}) = \mathsf{RT}(\emptyset\cup\{k\}, R\cup\{(k,v)\})$ because $R=\emptyset$. Let $K \neq \emptyset$ and let the statement hold for all sets smaller than K. Then $\mathsf{RT}(K,R) = (\mathsf{RT}(K_0,R_0),\mathsf{RT}(K_1,R_1);p,w)$ for some $w \in V\cup\{\bot\}$. Note that $p \neq k$ because $k \notin K$. Let $K' = K\cup\{k\}$ and $R' = R\cup\{(k,v)\}$. Then $\mathsf{gcp}(K') = \mathsf{gcp}(p,k)$.

If $p \prec k$, then $\mathsf{gcp}(K') = \mathsf{gcp}(K) = p$ and either $p\|0 \preceq k$ or $p\|1 \preceq k$.

- If $p\|0 \preceq k$ then $K'_0 = K_0\cup\{k\}$, $R'_0 = R_0\cup\{(k,v)\}$, $K'_1 = K_1$, $R'_1 = R_1$, and from $|K_0| < |K|$ it follows by induction that:

$$\begin{aligned}\mathsf{Add}(\mathsf{RT}(K,R);k,v) &= (\mathsf{Add}(\mathsf{RT}(K_0,R_0);k,v),\mathsf{RT}(K_1,R_1);p,w)\\ &= (\mathsf{RT}(K_0\cup\{k\},R_0\cup\{(k,v)\}),\mathsf{RT}(K_1,R_1);p,w)\\ &= (\mathsf{RT}(K'_0,R'_0),\mathsf{RT}(K'_1,R'_1);p,w) = \mathsf{RT}(K',R')\\ &= \mathsf{RT}(K\cup\{k\},R\cup\{(k,v)\})\end{aligned}$$

- If $p\|1 \preceq k$ then $K'_0 = K_0$, $R'_0 = R_0$, $K'_1 = K_1\cup\{k\}$, $R'_1 = R_1\cup\{(k,v)\}$ and from $|S_1| < |S|$ it follows by induction that:

$$\begin{aligned}\mathsf{Add}(\mathsf{RT}(K,R);k,v) &= (\mathsf{RT}(K_0,R_0),\mathsf{Add}(\mathsf{RT}(K_1,R_1);k,v);p,w)\\ &= (\mathsf{RT}(K_0,R_0),\mathsf{RT}(K_1\cup\{k\},R_1\cup\{(k,v)\});p,w)\\ &= (\mathsf{RT}(K'_0,R'_0),\mathsf{RT}(K'_1,R'_1);p,w) = \mathsf{RT}(K',R')\\ &= \mathsf{RT}(K\cup\{k\},R\cup\{(k,v)\})\end{aligned}$$

If $p \not\prec k$, then $|\mathsf{gcp}(p,k)| < |k|$ and hence, either $\mathsf{gcp}(p,k)\|0 \preceq k$ or $\mathsf{gcp}(p,k)\|1 \preceq k$.

- If $\mathsf{gcp}(p,k)\|0 \preceq k$ then $K'_0 = \{k\}$, $R'_0 = \{(k,v)\}$, $K'_1 = K$, $R'_1 = R$ and

$$\begin{aligned}\mathsf{Add}(\mathsf{RT}(K,R);k,v) &= ((\emptyset,\emptyset;k,v),\mathsf{RT}(K,R);\mathsf{gcp}(p,k),\bot)\\ &= (\mathsf{RT}(\{k\},\{(k,v)\}),\mathsf{RT}(K,R);\mathsf{gcp}(p,k),\bot)\\ &= (\mathsf{RT}(K'_0),\mathsf{RT}(K'_1);\mathsf{gcp}(K'),\bot) = \mathsf{RT}(K',R')\\ &= \mathsf{RT}(K\cup\{k\},R\cup\{(k,v)\})\end{aligned}$$

- If $\mathsf{gcp}(p,k)\|1 \preceq k$ then $K'_0 = K$, $R'_0 = R$, $K'_1 = \{k\}$, $R'_1 = \{(k,v)\}$ and

$$\begin{aligned}\mathsf{Add}(\mathsf{RT}(K,R);k,v) &= (\mathsf{RT}(K,R),(\emptyset,\emptyset;k,v);\mathsf{gcp}(p,k),\bot)\\ &= (\mathsf{RT}(K,R),\mathsf{RT}(\{k\},\{(k,v)\});\mathsf{gcp}(p,k),\bot)\\ &= (\mathsf{RT}(K'_0),\mathsf{RT}(K'_1);\mathsf{gcp}(K'),\bot) = \mathsf{RT}(K',R')\\ &= \mathsf{RT}(K\cup\{k\},R\cup\{(k,v)\})\end{aligned}$$

□

3 Tree Hash and Non-Inclusion Certificates

Definition 6 (Hash tree) *Let $H\colon \{0,1\}^* \to \{0,1\}^\kappa$ be a cryptographic hash function. The tree hash $\mathsf{trh}_H(T)$ of a tree T is computed by the following rules:*

1. *If $T = \emptyset$ then $\mathsf{trh}_H(T) = 0^\kappa$*
2. *If $T = (T_0, T_1; c)$ then $\mathsf{trh}_H(T) = H(\mathsf{trh}_H(T_0), \mathsf{trh}_H(T_1); c)$*

Definition 7 (Non-Inclusion certificate) *Let $K \subseteq \{0,1\}^\kappa$, $R\colon K \to V$, $k \in \{0,1\}^\kappa \backslash K$, $v \in V$, and $T = \mathsf{RT}(K, R)$. The non-inclusion certificate $\pi_{\mathsf{ni}}(T; k, v)$ is computed by the following rules:*

(1) $\pi_{\mathsf{ni}}(\emptyset; k, v) = []$ *and*

(2) if $T = (T_0, T_1; p, w)$ then:

- *(a)* $\pi_{\mathsf{ni}}(T; k, v) = (\mathsf{trh}_H(T_0), \mathsf{trh}_H(T_1); p, w)$ *if* $p \not\prec k$
- *(b)* $\pi_{\mathsf{ni}}(T; k, v) = (\mathsf{trh}_H(T_1); \mathsf{gcp}(T_1)) \| \pi_{\mathsf{ni}}(T_0; k, v)$ *if* $p\|0 \preceq k$
- *(c)* $\pi_{\mathsf{ni}}(T; k, v) = (\mathsf{trh}_H(T_0); \mathsf{gcp}(T_0)) \| \pi_{\mathsf{ni}}(T_1; k, v)$ *if* $p\|1 \preceq k$

Hence, every non-inclusion certificate π is either and empty list $[]$ or a list

$$[(h_1; p_1), \ldots, (h_m; p_m), (h^0, h^1; p, w)]$$

that begins with m pairs (m may be 0) and one single quadruple.

For example, if $T = (\emptyset, \emptyset; k', v')$ with $k \neq k' \in \{0,1\}^\kappa$ then $\pi_{\mathsf{ni}}(T; k, v) = (\mathsf{trh}_H(T); k') = (H(0^\kappa, 0^\kappa; k', v'), k')$.

Definition 8 (Previous output) *The previous output $\mathsf{out}_H(k; \pi)$ of a non-inclusion certificate π for a key k is defined by the following rules:*

(1) If $\pi = [(h_0, h_1; p, w)]$, $|p| \leq |k|$, and $p \not\preceq k$ then $\mathsf{out}_H(k; \pi) = (H(h_0, h_1; p, w), p)$. If $|p| > |k|$ or $p \preceq k$ then $\mathsf{out}_H(k; \pi) = \perp$.

(2) if $\pi = (h, p) \| \pi'$ with $\pi' \neq []$, $|p| \leq |k|$, $(r', q') = \mathsf{out}_H(k; \pi') \neq \perp$ and $\mathsf{gcp}(p, q') \prec k$ then

- *(a)* $\mathsf{out}_H(k; \pi) = (H(q', h; \mathsf{gcp}(p, p'), \perp), \mathsf{gcp}(p, q'))$, *if* $\mathsf{gcp}(p, q')\|0 \preceq k$
- *(b)* $\mathsf{out}_H(k; \pi) = (H(h, q'; \mathsf{gcp}(p, q'), \perp), \mathsf{gcp}(p, q'))$, *if* $\mathsf{gcp}(p, q')\|1 \preceq k$

(2') if $|p| > |k|$ or $\mathsf{out}_H(k; \pi') = \perp$ or $\mathsf{gcp}(p, q') \not\prec k$ in (2) then $\mathsf{out}_H(k; \pi) = \perp$.

Definition 9 (Next output) *The next output $\mathsf{out}'_H(k, v; \pi)$ of a non-inclusion certificate π for a key k is defined by the following rules:*

(1) If $\pi = [(h_0, h_1; p, w)]$, $|p| \le |k|$, *and* $p \not\preceq k$ *then with* $h = H(h_0, h_1; p, w)$*:*

(a) $\mathsf{out}'_H(k, v; \pi) = (H(H(0^\kappa, 0^\kappa; k, v), h; \mathsf{gcp}(p, k), \bot), \mathsf{gcp}(p, k))$ *if* $\mathsf{gcp}(p, k)\|0 \preceq k$

(b) $\mathsf{out}'_H(k, v; \pi) = (H(h, H(0^\kappa, 0^\kappa; k, v), \mathsf{gcp}(p, k), \bot), \mathsf{gcp}(p, k))$ *if* $\mathsf{gcp}(p, k)\|1 \preceq k$

(2) if $\pi = (h, p)\|\pi'$ *with* $\pi' \neq []$, $|p| \le |k|$, $(r'', q'') = \mathsf{out}'_H(k, v; \pi') \neq \bot$ *and* $\mathsf{gcp}(p, q'') \prec k$ *then*

(a) $\mathsf{out}'_H(k, v; \pi) = (H(r'', h; \mathsf{gcp}(p, q''), \bot), \mathsf{gcp}(p, q''))$, *if* $\mathsf{gcp}(p, q'')\|0 \preceq k$

(b) $\mathsf{out}'_H(k, v; \pi) = (H(h, r''; \mathsf{gcp}(p, q''), \bot), \mathsf{gcp}(p, q''))$, *if* $\mathsf{gcp}(p, q'')\|1 \preceq k$

(2') if $|p| > |k|$ *or* $\mathsf{out}'_H(k, v; \pi') = \bot$ *or* $\mathsf{gcp}(p, q'') \not\preceq k$ *in (2) then* $\mathsf{out}'_H(k; \pi) = \bot$.

Theorem 1 *If* $\mathsf{out}_H(k; \pi) = (\mathsf{trh}_H(T), \mathsf{gcp}(T)) \neq \bot$, $T = \mathsf{RT}(K, R)$, *and no* H*-collisions can be extracted from the computations of* $\mathsf{out}_H(k; \pi)$ *and* $\mathsf{trh}_H(T)$*, then*

$$\mathsf{out}'_H(k, v; \pi) = (\mathsf{trh}_H(\mathsf{Add}(T; k, v)), \mathsf{gcp}(\mathsf{Add}(T; k, v))) \ .$$

Proof: Assume that no collisions occur in computations $(r, q) \leftarrow \mathsf{out}_H(k; \pi)$, $r \leftarrow \mathsf{trh}_H(T)$. Let $T = (T_0, T_1; \mathsf{gcp}(K), w)$, where $T_0 = \mathsf{RT}(K_0, R_0)$, $T_1 = \mathsf{RT}(K_1, R_1)$. From $r = \mathsf{trh}_H(T)$ it follows that $r = H(r_0, r_1; \mathsf{gcp}(K), w)$, where $r_0 = \mathsf{trh}_H(T_0)$ and $r_1 = \mathsf{trh}_H(T_1)$. We use induction on the length of π.

(1) If $\pi = [(h_0, h_1, p, w)]$ then from $\mathsf{out}_H(k; \pi) \neq \bot$ it follows that $|p| \le |k|$ and $p \not\preceq k$. Hence, either $\mathsf{gcp}(p, k)\|0 \preceq k$ or $\mathsf{gcp}(p, k)\|1 \preceq k$. Let $h = H(h_0, h_1, p, w)$. It follows from the definition of out_H that $r = h = \mathsf{trh}_H(T)$, $q = p = \mathsf{gcp}(T) = \mathsf{gcp}(K)$, and $\mathsf{gcp}(p, k) = \mathsf{gcp}(\mathsf{Add}(T; k, v))$.

1. If $\mathsf{gcp}(p, k)\|0 \preceq k$ then by definitions of out'_H, trh_H and Add:

$$\begin{aligned}
\mathsf{out}'_H(k, v; \pi) &= (H(H(0^\kappa, 0^\kappa; k, v), h; \mathsf{gcp}(p, k), \bot), \mathsf{gcp}(p, k)) \\
&= (\mathsf{trh}_H(((\emptyset, \emptyset; k, v), T; \mathsf{gcp}(p, k), \bot)), \mathsf{gcp}(p, k)) \\
&= (\mathsf{trh}_H(\mathsf{Add}(T; k, v)), \mathsf{gcp}(p, k)) \\
&= (\mathsf{trh}_H(\mathsf{Add}(T; k, v)), \mathsf{gcp}(\mathsf{Add}(T; k, v))) \ .
\end{aligned}$$

2. If $\mathsf{gcp}(p, k)\|1 \preceq k$ then:

$$\begin{aligned}
\mathsf{out}'_H(k, v; \pi) &= (\mathsf{trh}_H((T, (\emptyset, \emptyset; k, v); \mathsf{gcp}(p, k), \bot)), \mathsf{gcp}(p, k)) \\
&= (\mathsf{trh}_H(\mathsf{Add}(T; k, v)), \mathsf{gcp}(p, k)) \\
&= (\mathsf{trh}_H(\mathsf{Add}(T; k, v)), \mathsf{gcp}(\mathsf{Add}(T; k, v))) \ .
\end{aligned}$$

(2) Let $\pi = (h, p)\|\pi'$ with $\pi' \neq []$ and the statement hold for all certificates shorter than π. From $\mathsf{out}_H(k; \pi) \neq \bot$ it follows that $|p| \le |k|$, $(r', q') = \mathsf{out}_H(k; \pi') \neq \bot$ and $\mathsf{gcp}(p, q') \prec k$. Hence, either $\mathsf{gcp}(p, q')\|0 \preceq k$ or $\mathsf{gcp}(p, q')\|1 \preceq k$.

1. If $\mathsf{gcp}(p,q')\|0 \preceq k$ then

$$\begin{array}{rcl}(H(r',h;\mathsf{gcp}(p,q'),\bot),\mathsf{gcp}(p,q')) & = & \mathsf{out}_H(k;\pi) = (r,\mathsf{gcp}(p,q')) = (\mathsf{trh}_H(T),\mathsf{gcp}(p,q')) \\ & = & (H(r_0,r_1;\mathsf{gcp}(K),w),\mathsf{gcp}(p,q'))\end{array}$$

Hence $H(r',h;\mathsf{gcp}(p,q'),\bot) = H(r_0,r_1;\mathsf{gcp}(K),w)$ and as there are no collisions, $r_0 = r'$, $r_1 = h$, $\mathsf{gcp}(K) = \mathsf{gcp}(p,q')$, and $w = \bot$, implying $r' = \mathsf{trh}_H(T_0)$ with $T_0 = \mathsf{RT}(K_0,R_0)$. In order to apply the induction assumption we it remains to show that $q' = \mathsf{gcp}(T_0)$. Indeed,

   (a) If $\pi' = [(h'_0,h'_1,p',w')]$ then $r' = H(h'_0,h'_1,p',w')$ and $q' = p'$. But as $r' = r_0 = \mathsf{trh}_H(T_0) = H(h''_0,h''_1;\mathsf{gcp}(T_0),w'')$ for some h''_0,h''_1,w'' then from no collisions assumption it follows that $q' = p' = \mathsf{gcp}(T_0)$.

   (b) If $\pi' = (h',p')\|\pi''$ where $\pi'' \neq []$, then $(r',q') = \mathsf{out}_H(k;\pi') = (H(x_0,x_1;q',y),q')$ for some x_0,x_1,y but also $r' = r_0 = \mathsf{trh}_H(T_0) = H(h''_0,h''_1;\mathsf{gcp}(T_0),w'')$ for some h''_0,h''_1,w'' then from no collisions assumption it follows that $q' = \mathsf{gcp}(T_0)$.

   Therefore, by induction assumption:

$$\mathsf{out}'_H(k,v;\pi') = (\mathsf{trh}_H(\mathsf{Add}(T_0;k,v)),\mathsf{gcp}(\mathsf{Add}(T_0;k,v))) \neq \bot \ , \tag{1}$$

   because there cannot be a collision in the computations of $\mathsf{out}_H(k;\pi)$ and $\mathsf{trh}_H(T)$. Indeed, the computation of $\mathsf{out}_H(k;\pi)$ involves the computation of $\mathsf{out}_H(k;\pi')$ and the computation of $\mathsf{trh}_H(T)$ involves the computation of $\mathsf{trh}_H(T_0)$. Therefore, by the definition of out'_H and equation (1):

$$\mathsf{out}'_H(k,v;\pi) = (H(r'',h;\mathsf{gcp}(p,q''),\bot),\mathsf{gcp}(p,q'')) \tag{2}$$

   where $(r'',q'') = \mathsf{out}'_H(k,v;\pi')$ and hence, by (1), $r'' = \mathsf{trh}_H(\mathsf{Add}(T_0;k,v))$ and $q'' = \mathsf{gcp}(q',k)$. Therefore, as $h = \mathsf{trh}_H(T_1)$ and $\mathsf{gcp}(K)\|0 = \mathsf{gcp}(p,q')\|0 \preceq k$, we have $\mathsf{Add}(T;k,v) = (\mathsf{Add}(T_0;k,v),T_1;\mathsf{gcp}(K),w)$ and

$$\begin{array}{rcl}\mathsf{out}'_H(k,v;\pi) & = & (H(\mathsf{trh}_H(\mathsf{Add}(T_0;k,v)),\mathsf{trh}_H(T_1);\mathsf{gcp}(p,q''),\bot),\mathsf{gcp}(p,q'')) \\ & = & (H(\mathsf{trh}_H(\mathsf{Add}(T_0;k,v)),\mathsf{trh}_H(T_1);\mathsf{gcp}(K\cup\{k\}),\bot),\mathsf{gcp}(K\cup\{k\})) \\ & = & (\mathsf{Add}(T;k,v),\mathsf{gcp}(\mathsf{Add}(T;k,v)))\ ,\end{array}$$

   because $\mathsf{gcp}(p,q'') = \mathsf{gcp}(p,\mathsf{gcp}(q',k)) = \mathsf{gcp}(\mathsf{gcp}(p,q'),k) = \mathsf{gcp}(\mathsf{gcp}(K),k) = \mathsf{gcp}(K\cup\{k\})$

2. The case $\mathsf{gcp}(p,q')\|1 \preceq k$ is proved analogously.

□